

Experiment 10: A* Algorithm Aim:

Find the shortest path.

Algorithm:

1. Start.
2. Apply the required AI technique.
3. Generate the solution.
4. Display the result.
5. Stop.

Python Code: graph =

```
{  
  
'A': [('B', 1), ('C', 3)],  
  
'B': [('D', 3), ('E', 1)],  
  
'C': [('F', 5)],  
  
'D': [],  
  
'E': [('F', 1)],  
  
'F': []  
  
}  
  
heuristic = {  
  
'A': 6, 'B': 4,  
  
'C': 2, 'D': 3,  
  
'E': 1,  
  
'F': 0  
  
}  
  
open_list = [('A', 0)]  
  
visited = []  
  
while open_list:  
  
    open_list.sort(key=lambda x: x[1] + heuristic[x[0]])  
  
    node, cost = open_list.pop(0)  
  
    if node in visited:  
  
        continue  
  
    print(node)
```

```
visited.append(node)

if node=='F':    print("Goal
Reached")    break

for n,w in graph[node]:
    open_list.append((n,cost+w))
```

Output

```
Visited: A
Visited: B
Visited: E
Visited: F
Goal Reached
Total Cost: 3
,
```

Result:

Goal reached.